

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

FORMAL DIGITAL FORENSICS INVESTIGATION REPORT

SBT-DF203 - LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Field	Detail
Case Identifier	SBT-DF203-Lab9
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Registration / ID No.	202520850LE
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Date of Submission	September 26, 2026
Primary Target System	Kali Linux Workstation (kali@kali)
Evidence Type	Historical WEP40-encrypted 802.11 PCAP capture (CodeGate CTF 2015) and associated analysis artifacts

Prepared for authorized academic forensic examination

1. Executive Summary

This report documents the completion of SBT-DF203 Lab 9, a WEP40 wireless-forensics exercise conducted entirely offline in an isolated Kali Linux analysis environment using a historical, publicly available training capture (CodeGate CTF 2015, "Good Crypto" challenge). The practical established the lab workspace, preserved and hashed the supplied evidence, inventoried IEEE 802.11 frame types, identified WEP-protected data frames and repeated initialization vectors (IVs), recovered the 40-bit WEP key using Aircrack-ng, decrypted the capture offline with airdecap-ng, extracted network endpoints and conversations, and carved embedded images, HTML, and other objects from the decrypted traffic.

The target network was identified as BSSID 00:26:66:55:97:D6, ESSID "cgnetwork", secured with WEP encryption across 45,169 total captured frames, of which 15,477 were WEP-protected data frames. Aircrack-ng recovered the 40-bit WEP key A4:3D:F6:F3:74 with 100% decryption confidence after testing 83 candidate keys against the captured IVs. The key was applied with airdecap-ng, successfully decrypting all 15,477 WEP data packets with zero corrupted frames, producing the working file file_working-dec.

Post-decryption analysis identified the host device as 192.168.0.15 (MAC f0:f6:1c:68:96:7c, an Apple device) and a second local host at 192.168.0.9 communicating over TCP port 5000. The gateway/router was identified at 192.168.0.1 (MAC 00:26:66:55:97:d4). Reconstructed traffic showed the host browsing Reddit and Imgur (specifically the "Epic Dogpile" gallery post), generating substantial HTTP, DNS, TLS, and third-party analytics/advertising traffic. Object carving via foremost and tshark HTTP export recovered 17 foremost-carved files (JPG, GIF, PNG, HTM) and approximately 200 additional objects (images, JavaScript, CSS, fonts, and JSON) directly from reconstructed HTTP sessions.

The optional Part G (historical CTF passphrase-cracking challenge) was not performed as part of this submission, consistent with the lab instructions identifying it as instructor-assessed and optional.

The supplied evidence/file.xz and its decompressed working copy working/file_working matched at SHA-256 c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d. The decrypted capture working/file_working-dec was hashed at SHA-256 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04.

2. Investigation Overview and Scope

2.1 Purpose and Objectives

- Preserve and hash the supplied WEP-encrypted evidence before analysis.
- Inventory IEEE 802.11 frame types (Management, Control, Data) present in the capture.
- Identify WEP-protected data frames, the target BSSID/ESSID, and evidence of IV reuse.
- Recover or validate the 40-bit WEP key using Aircrack-ng.
- Decrypt the capture offline using airdecap-ng and verify successful decryption.
- Extract Ethernet and IP endpoints, TCP conversations, and IP-to-MAC mappings from decrypted traffic.
- Carve and inventory images, HTML, and other transferred objects from the decrypted capture.

- Document the cryptographic weaknesses of WEP/RC4 that permitted key recovery.
- Document limitations and operational corrections without treating them as anomalies.

2.2 Environment and Tools

Component	Observed configuration
Operating System	Kali Linux (kali@kali)
Analysis directory	~/SBT-DF203-Lab9
Evidence source	Historical CodeGate CTF 2015 "Good Crypto" capture (file.xz)
Target BSSID / ESSID	00:26:66:55:97:D6 / cgnetwork
Encryption observed	WEP (15,477 IVs captured)
Key-recovery tool	aircrack-ng 1.7
Decryption tool	airdecap-ng
Packet analysis tools	tshark / Wireshark
Object carving tools	foremost, tshark --export-objects

2.3 Legal, Ethical and Operational Boundaries

All analysis in this lab was performed exclusively offline against a single historical, publicly distributed training capture. No live wireless network was scanned, targeted, or connected to at any point. The recovered WEP key and all analysis outputs are confined to this coursework submission and were not applied to any production, campus, or third-party network.

3. Evidence Preservation and Integrity

The lab workspace was created under ~/SBT-DF203-Lab9 with subdirectories evidence/, working/, exported/, reports/, screenshots/, and scripts/, following the required evidence-handling structure.

```

kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(kali@kali)~$ sudo apt update
[sudo] password for kali:
Get:2 https://repo.protonvpn.com/debian stable InRelease [2,967 B]
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:4 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:8 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 k
B]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16
.0 kB]
Fetched 77.3 MB in 5min 55s (218 kB/s)
1921 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
binwalk is already the newest version (2.4.3+dfsg1-3).
binwalk set to manually installed.
The following packages were automatically installed and are no longer required:
libimath-3-1-29t64 libpython3.13-dev python3.13 rpcsvc-proto
libmbedcrypto16 libraw23t64 python3.13-dev
libpostal-data libvidstab1.1 python3.13-minimal
libpostal1 libximg3 python3.13-venv
Use 'sudo apt autoremove' to remove them.

```

Figure 1: Lab 9 workspace tooling verification - aircrack-ng, wireshark, tshark, foremost, and binwalk confirmed installed.

```

Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab9
(kali@kali)-[~/SBT-DF203-Lab9]
$ pwd
/home/kali/SBT-DF203-Lab9
(kali@kali)-[~/SBT-DF203-Lab9]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

```

Figure 2: Lab 9 workspace directories created under ~/SBT-DF203-Lab9.

The supplied capture was downloaded from the CodeGate CTF 2015 GitHub write-up repository directly into evidence/file.xz and hashed immediately upon download.

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ wget -O evidence/file.xz \
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/p
rogramming/good-crypto/file.xz'
--2026-09-26 09:22:50-- https://raw.githubusercontent.com/ctfs/write-ups-2015/ma
ster/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.108.13
3, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.1
33|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz  100%[=====>] 12.24M  790KB/s  in 18s
2026-09-26 09:23:09 (707 KB/s) - 'evidence/file.xz' saved [12837660/12837660]

(kali@kali)-[~/SBT-DF203-Lab9]
$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.x
z
(kali@kali)-[~/SBT-DF203-Lab9]
$

```

Figure 3: Supplied evidence/file.xz downloaded and hashed (SHA-256 dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6).

The evidence file was copied to a working directory, decompressed, and both the compressed and decompressed files were re-hashed to confirm bit-for-bit integrity throughout the preservation workflow.

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ cp --preserve=timestamps evidence/file.xz working/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$ unxz -k working/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$ file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version
2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203-Lab9]
$ sha256sum working/file_working.xz working/file_working | tee
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_wo
rking.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_wo
rking

(kali@kali)-[~/SBT-DF203-Lab9]
$ sha256sum working/file_working.xz working/file_working | tee reports/working_
hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_wo
rking.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_wo
rking

(kali@kali)-[~/SBT-DF203-Lab9]
$

```

Figure 4: Evidence decompression, file-type identification (IEEE 802.11 pcap, version 2.4), and dual hashing of the compressed and working capture files.

3.1 Evidence and Chain-of-Custody Summary

Evidence / Artifact	Role	SHA-256
evidence/file.xz	Supplied/original compressed evidence	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
working/file_working.xz	Working copy of compressed evidence	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
working/file_working	Decompressed working capture (IEEE 802.11)	c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
working/file_working-dec	Decrypted capture (post airdecap-ng)	167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04

4. Part A - Inventory of the Wireless Capture

capinfos and tshark protocol-hierarchy statistics were used to characterize the overall capture, followed by a field-level extraction of the first 100 wlan frames to classify frame categories.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ grep -m 5 -P '\t0\t' reports/wlan_frame_sample.tsv
54      2015-03-06T02:51:54.367633000-0500      8      00:26:66:55:97:d6
ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6

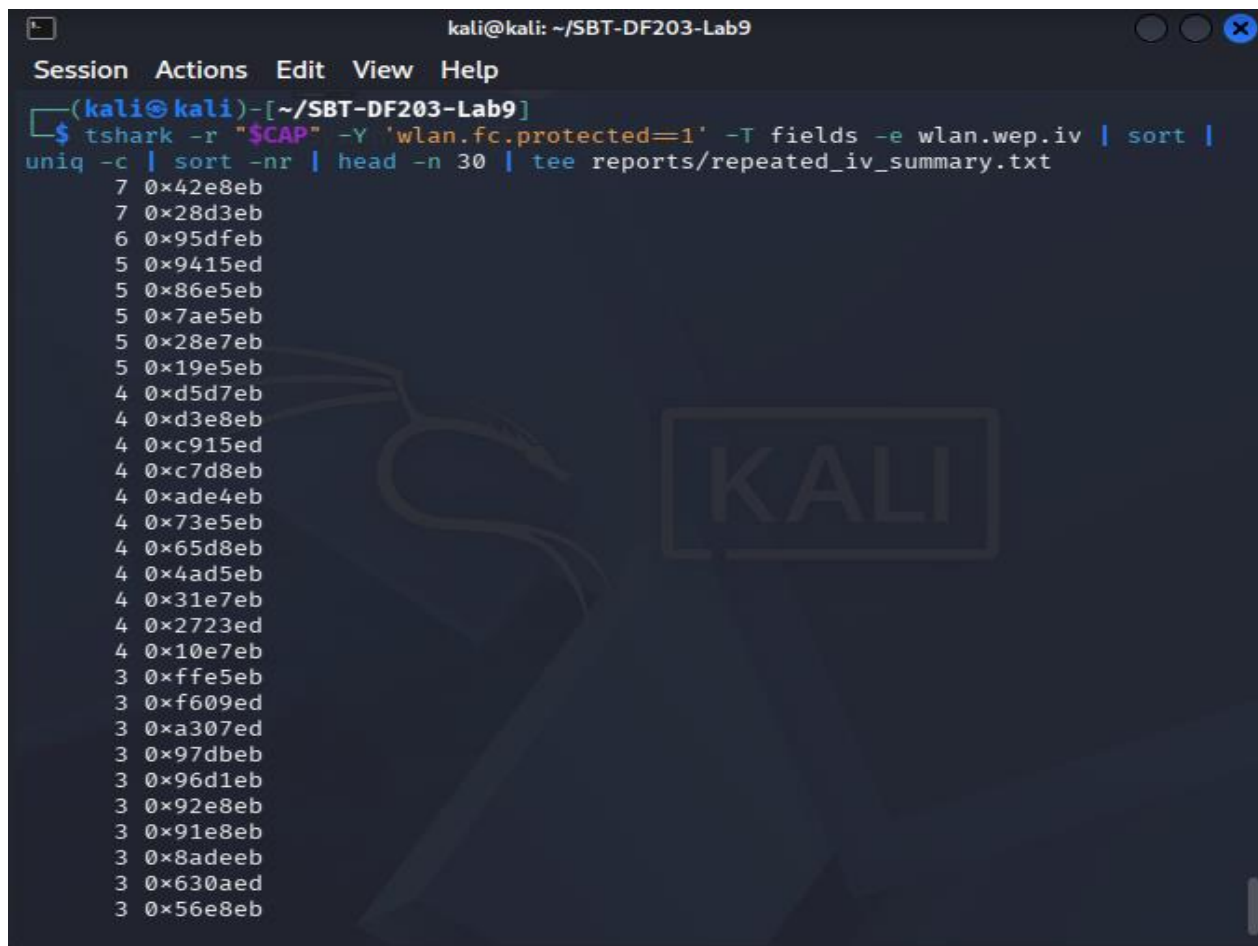
(kali@kali)-[~/SBT-DF203-Lab9]
$
```

Figure 5: capinfos output - 45,169 packets, 272.97-second duration, IEEE 802.11 Wireless LAN encapsulation, SHA256 c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d confirming no drift from the working file.

```
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e frame.number -e frame
.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key |
head -n 200 | tee reports/wep_protected_frames.tsv
9432      1425628319.232982000      00:26:66:55:97:d4      ff:ff:ff:ff:ff:ff      0
0:26:66:55:97:d6      0x67ffec      0
10130     1425628320.138252000      f0:f6:1c:68:96:7c      00:26:66:55:97:d6      0
0:26:66:55:97:d6      0xc2cbeb      0
10155     1425628320.176656000      f0:f6:1c:68:96:7c      04:8d:38:48:e5:b4      0
0:26:66:55:97:d6      0xc3cbeb      0
10157     1425628320.177168000      f0:f6:1c:68:96:7c      04:8d:38:48:a8:b5      0
0:26:66:55:97:d6      0xc4cbeb      0
10159     1425628320.177168000      f0:f6:1c:68:96:7c      08:10:77:92:7c:2f      0
0:26:66:55:97:d6      0xc5cbeb      0
10161     1425628320.177168000      f0:f6:1c:68:96:7c      00:26:66:55:97:d4      0
0:26:66:55:97:d6      0xc6cbeb      0
10163     1425628320.177168000      f0:f6:1c:68:96:7c      c8:3a:35:58:c4:ba      0
0:26:66:55:97:d6      0xc7cbeb      0
10165     1425628320.177167000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      0
0:26:66:55:97:d6      0xc8cbeb      0
10167     1425628320.177686000      00:26:66:55:97:d4      f0:f6:1c:68:96:7c      0
0:26:66:55:97:d6      0x68ffec      0
10170     1425628320.180760000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      0
0:26:66:55:97:d6      0x69ffec      0
10171     1425628320.182288000      f0:f6:1c:68:96:7c      33:33:00:00:00:02      0
0:26:66:55:97:d6      0xc9cbeb      0
10172     1425628320.187417000      00:26:66:55:97:d4      ff:ff:ff:ff:ff:ff      0
0:26:66:55:97:d6      0x6affec      0
10173     1425628320.187408000      f0:f6:1c:68:96:7c      33:33:00:00:00:02      0
0:26:66:55:97:d6      0xc9cbeb      0
10175     1425628320.187919000      f0:f6:1c:68:96:7c      ff:ff:ff:ff:ff:ff      0
```

Figure 6: Protocol hierarchy statistics - all 45,169 frames classified as wlan; 15,713 identified as data frames.



```

kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv | sort |
uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
 7 0x42e8eb
 7 0x28d3eb
 6 0x95dfef
 5 0x9415ed
 5 0x86e5eb
 5 0x7ae5eb
 5 0x28e7eb
 5 0x19e5eb
 4 0xd5d7eb
 4 0xd3e8eb
 4 0xc915ed
 4 0xc7d8eb
 4 0xade4eb
 4 0x73e5eb
 4 0x65d8eb
 4 0x4ad5eb
 4 0x31e7eb
 4 0x2723ed
 4 0x10e7eb
 3 0xffe5eb
 3 0xf609ed
 3 0xa307ed
 3 0x97dbeb
 3 0x96d1eb
 3 0x92e8eb
 3 0x91e8eb
 3 0x8adeeb
 3 0x630aed
 3 0x56e8eb

```

Figure 7: Field extraction of the first wlan frames, showing frame.type/subtype, source, destination, and BSSID.

4.1 Frame Category Classification

Frame Category	Type Value	Purpose	Example from Capture
Management	0	Beacons, authentication, association	Frame #54, subtype 8 (Beacon), src 00:26:66:55:97:d6, dst ff:ff:ff:ff:ff:ff
Control	1	Flow control and acknowledgements	Frame #1, subtype 11 (Request-to-Send)
Data	2	Carries network payload	15,713 frames observed in protocol hierarchy

5. Part B - Identification of WEP Protection and IV Evidence

Data frames with the protected bit set (`wlan.fc.protected==1`) were filtered to confirm WEP encryption and extract per-frame initialization vector (IV) values. A frequency count of IV values within a 200-frame sample was then produced to demonstrate IV reuse.

```

Session  Actions  Edit  View  Help

(kali@kali)-[~/SBT-DF203-Lab9]
$ aircrack-ng working/file_working | tee reports/aircrack_output.txt
Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

#  BSSID          ESSID          Encryption
1  00:26:66:55:97:D6  cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB    depth  byte(vote)
0     0/ 1   A4(22784) 62(20992) A8(19968) B6(19968) 42(19456)
1     0/ 1   3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968)
2     0/ 1   F6(23808) E4(20992) D0(20736) 68(20224) 95(19712)
3     1/ 10  F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456)
4     6/ 9   01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456)

```

Figure 8: Protected (WEP) data frames filtered from the capture. All protected frames share BSSID 00:26:66:55:97:d6; wlan.wep.key (key index) is 0 for every sampled frame, indicating a single active WEP key slot.

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ cd working

(kali@kali)-[~/SBT-DF203-Lab9/working]
$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

```

Figure 9: Repeated IV frequency summary - IV 0x42e8eb recurs 7 times within the 200-frame sample; several other IVs recur 4-6 times.

64-bit WEP (WEP-40) combines a 24-bit initialization vector with a 40-bit secret key to seed the RC4 keystream generator. Because the IV space is only 2^{24} (approximately 16.7 million) values, busy networks exhaust the space quickly. Two packets encrypted under an identical IV use an identical RC4 keystream; XOR-ing their ciphertexts cancels the keystream and yields the XOR of the two plaintexts, which can be exploited to recover plaintext statistically. This capture, with 15,477 WEP data packets from a single BSSID, provided ample IV reuse for aircrack-ng's statistical key-recovery attack, consistent with the theoretical basis presented in the accompanying lecture module.

6. Part C - WEP40 Key Recovery

Aircrack-ng was executed directly against the working capture file with no supplied key, to independently recover the 40-bit WEP key from the captured IVs.

```

Session  Actions  Edit  View  Help

(kali@kali)-[~/SBT-DF203-Lab9]
$ aircrack-ng working/file_working | tee reports/aircrack_output.txt
Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

#  BSSID          ESSID          Encryption
1  00:26:66:55:97:D6  cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB    depth  byte(vote)
0     0/ 1    A4(22784) 62(20992) A8(19968) B6(19968) 42(19456)
1     0/ 1    3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968)
2     0/ 1    F6(23808) E4(20992) D0(20736) 68(20224) 95(19712)
3     1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456)
4     6/ 9     01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456)

```

Figure 10: aircrack-ng identifying the target network (BSSID 00:26:66:55:97:D6, ESSID cgnetwork, WEP with 15,477 IVs) prior to key recovery.

The full key-recovery attempt returned the following result:

KEY FOUND! [A4:3D:F6:F3:74] Decrypted correctly: 100%

6.1 Key Recovery Summary

Field	Value
BSSID	00:26:66:55:97:D6
ESSID	cgnetwork
Encryption	WEP, 15,477 IVs captured
Candidate keys tested	83
Key length	40 bits (5 bytes)
Recovered key	A4:3D:F6:F3:74
Decryption confidence	100% correct

The near-instantaneous convergence of the attack (83 tested candidates against 15,477 IVs) is consistent with the reduced effective key-schedule entropy of WEP-40 described in the accompanying lecture module - the small IV space combined with RC4's statistical keystream biases allows practical key recovery well before the full 2^{40} keyspace would otherwise need to be searched.

7. Part D - Offline Decryption

The recovered key was applied using `airdecap-ng` in the hyphen-free hexadecimal format required by the tool (A43DF6F374, with colons removed).

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ cd working

(kali@kali)-[~/SBT-DF203-Lab9/working]
$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets     0
Number of decrypted WPA packets     0
Number of bad TKIP (WPA) packets    0
Number of bad CCMP (WPA) packets    0
```

Figure 11: airdecap-ng decryption run - 15,477 of 15,477 WEP data packets decrypted successfully with 0 corrupted frames and 0 WPA/TKIP/CCMP errors.

```

(kali㉿kali)-[~/SBT-DF203-Lab9/working]
$ cd ..

(kali㉿kali)-[~/SBT-DF203-Lab9]
$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.
txt
 1574837  13904 -rw-rw-r--  1 kali    kali    14234002 Sep 26 09:23 working/f
ile_working
 1574851  12488 -rw-rw-r--  1 kali    kali    12784501 Sep 26 09:49 working/f
ile_working-dec
 1574836  12540 -rw-rw-r--  1 kali    kali    12837660 Sep 26 09:23 working/f
ile_working.xz

(kali㉿kali)-[~/SBT-DF203-Lab9]
$ sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_wo
rking
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_wo
rking-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_wo
rking.xz

(kali㉿kali)-[~/SBT-DF203-Lab9]
$

```

Figure 12: Post-decryption file inventory and SHA-256 hashing of file_working, file_working.xz, and the newly created file_working-dec.

Metric	Value
Total stations seen	10
Total packets read	45,169
Total WEP data packets	15,477
Decrypted WEP packets	15,477 (100%)
Corrupted WEP packets	0
Decrypted capture filename	working/file_working-dec
Decrypted capture SHA-256	167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04

8. Part E - Endpoint and Conversation Analysis

With the decrypted capture available, tshark endpoint and conversation statistics were generated to identify host/server IP and MAC addresses, and a field extraction correlated 802.11 source/destination MACs against IP-layer traffic.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ DEC=working/file_working-dec

(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
```

```
Ethernet Endpoints
Filter:<No Filter>
```

	Packets	Bytes	Tx Packets	Tx Bytes	Rx P
Apple_68:96:7c	15407	12 MB	7076	616 kB	
8331		11 MB			
EFMNetworks_55:97:d4	15020	12 MB	8178	11 MB	
6842		593 kB			
ASUSTekCOMPU_2a:c2:7a	358	281 kB	223	270 kB	
135		11 kB			
IPv4mcast_fb	42	6,648 bytes	0	0 bytes	
42		6,648 bytes			
IPv6mcast_fb	36	7,022 bytes	0	0 bytes	
36		7,022 bytes			
Broadcast	31	3,386 bytes	0	0 bytes	
31		3,386 bytes			
IPv4mcast_7f:ff:fa	31	11 kB	0	0 bytes	
31		11 kB			
IPv6mcast_16	10	920 bytes	0	0 bytes	
10		920 bytes			
IPv6mcast_02	9	630 bytes	0	0 bytes	
9		630 bytes			
IPv4mcast_16	3	180 bytes	0	0 bytes	
3		180 bytes			
IPv4mcast_02	2	92 bytes	0	0 bytes	

Figure 13: Ethernet endpoint statistics from the decrypted capture - Apple_68:96:7c (host, 15,407 packets) and EFMNetworks_55:97:d4 (router, 15,020 packets) dominate the traffic.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
```

```
IPv4 Endpoints
Filter:<No Filter>
```

	Packets	Bytes	Tx Packets	Tx Bytes	Rx P
192.168.0.15	15309	12 MB	6995	608 kB	
8314		11 MB			
198.90.20.111	8057	6,916 kB	4446	6,637 kB	
3611		279 kB			
199.27.79.193	5694	4,825 kB	3131	4,603 kB	
2563		221 kB			
192.168.0.9	329	277 kB	197	265 kB	
132		11 kB			
173.194.127.141	153	109 kB	80	102 kB	
73		7,364 bytes			
173.194.127.55	143	89 kB	71	82 kB	
72		6,990 bytes			
8.8.8.8	81	10 kB	40	6,960 bytes	
41		3,391 bytes			
199.16.158.186	66	15 kB	29	10 kB	
37		5,214 bytes			
173.194.127.227	66	24 kB	26	15 kB	
40		8,455 bytes			
173.194.127.229	55	22 kB	23	13 kB	
32		9,731 bytes			
31.13.70.1	51	19 kB	26	15 kB	
25		3,151 bytes			
173.241.248.220	44	6,848 bytes	20	3,122 bytes	
24		3,726 bytes			
198.41.208.140	44	28 kB	21	25 kB	

Figure 14: IPv4 endpoint statistics - host 192.168.0.15 (15,309 packets) and local server 192.168.0.9 (329 packets) alongside external web servers 198.90.20.111 and 199.27.79.193.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
```

TCP Conversations
Filter:<No Filter>

→	Total	Relative	Duration	←	Frames	Bytes	Fr
192.168.0.15:51007 ↔ 198.90.20.111:80	2235 1,958 kB	113.588792000	45.4035	1257 1,885 kB	9		
192.168.0.15:50959 ↔ 199.27.79.193:80	2168 1,887 kB	21.074236000	34.8633	1209 1,816 kB	9		
192.168.0.15:51008 ↔ 198.90.20.111:80	2106 1,810 kB	113.588792000	46.0179	1159 1,739 kB	9		
192.168.0.15:51001 ↔ 199.27.79.193:80	2072 1,791 kB	63.066042000	26.6088	1145 1,725 kB	9		
192.168.0.15:51009 ↔ 198.90.20.111:80	1259 1,072 kB	113.605175000	28.7983	687 1,028 kB	5		
192.168.0.15:51014 ↔ 198.90.20.111:80	1230 1,036 kB	114.020474000	25.0538	664 991 kB	5		
192.168.0.15:51006 ↔ 198.90.20.111:80	1174 1,011 kB	113.578039000	25.4942	654 970 kB	5		
192.168.0.15:51000 ↔ 199.27.79.193:80	698 586 kB	58.731642000	4.4552	375 561 kB	3		
192.168.0.15:50999 ↔ 199.27.79.193:80	386 319 kB	56.792636000	1.9390	206 303 kB	1		
192.168.0.15:51036 ↔ 192.168.0.9:5000	168 150 kB	243.325633000	0.1777	98 144 kB			
192.168.0.15:50958 ↔ 199.27.79.193:80	140 105 kB	20.145401000	14.9846	73 93 kB			
192.168.0.15:51035 ↔ 192.168.0.9:5000				80 116 kB			

Figure 15: TCP conversation statistics showing the busiest sessions between 192.168.0.15 and external HTTP servers on port 80, plus a distinct local session to 192.168.0.9:5000.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.p.src -e ip.dst -e _ws.col.Protocol | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```

7	0.0.0.0	255.255.255.255	DHCP
9	0.0.0.0	255.255.255.255	DHCP
11	192.168.0.1	255.255.255.255	DHCP
20	192.168.0.15	8.8.8.8	DNS
21	192.168.0.15	8.8.8.8	DNS
22	192.168.0.15	224.0.0.2	IGMPv2
23	192.168.0.15	224.0.0.251	IGMPv2
24	192.168.0.15	224.0.0.2	IGMPv2
25	192.168.0.15	224.0.0.251	IGMPv2
26	8.8.8.8	192.168.0.15	DNS
27	8.8.8.8	192.168.0.15	DNS
30	192.168.0.15	17.172.239.31	TCP
31	192.168.0.15	17.172.239.127	TCP
33	17.172.239.31	192.168.0.15	TCP
35	192.168.0.15	17.172.239.31	TCP
36	192.168.0.15	17.172.239.31	TLSv1
37	17.172.239.127	192.168.0.15	TCP
38	192.168.0.15	17.172.239.127	TCP
42	17.172.239.31	192.168.0.15	TCP
43	17.172.239.31	192.168.0.15	TLSv1
44	17.172.239.31	192.168.0.15	TLSv1
45	17.172.239.31	192.168.0.15	TLSv1
47	0.0.0.0	255.255.255.255	DHCP
48	0.0.0.0	255.255.255.255	DHCP
49	192.168.0.15	17.172.239.31	TCP
50	192.168.0.15	17.172.239.31	TCP
51	192.168.0.1	192.168.0.15	DHCP
52	192.168.0.15	17.172.239.31	TLSv1

Figure 16: IP/MAC mapping sample - DHCP lease of 192.168.0.15 from gateway 192.168.0.1, followed by DNS, TCP, and TLS sessions to external hosts.

8.1 Identified Endpoints

Role	IP Address	MAC Address	Notes
Host	192.168.0.15	f0:f6:1c:68:96:7c	Apple device; dominant local traffic source (15,309 IP packets)
Local server	192.168.0.9	ASUSTekCOMPU_2a:c2:7a	Communicates with host over non-standard TCP port 5000
Gateway / Router	192.168.0.1	00:26:66:55:97:d4	Issues DHCP lease to host; matches BSSID observed during WEP key recovery
External web servers	198.90.20.111 / 199.27.79.193	N/A (routed)	Heaviest external HTTP traffic (imgur.com, associated CDN)

8.2 Protocols Observed

DHCP, ARP, DNS, HTTP, TLSv1 / TLSv1.2, IGMPv2, MDNS, and ICMPv6 were all observed within the decrypted traffic, consistent with a typical consumer device browsing session.

9. Part F - Extraction of Images, HTML, and Other Objects

Two independent carving/extraction methods were applied to the decrypted capture: tshark's native HTTP object exporter (reconstructing files from complete HTTP request/response pairs) and foremost (signature-based file carving across the entire capture, independent of protocol reconstruction).

```
(kali@kali)~[~/SBT-DF203-Lab9]
$ mkdir -p exported/http_objects exported/foremost

(kali@kali)~[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
  1  0.000000 EFMNetworks_55:97:d4 → Broadcast      ARP 42 Who has 192.168.0.15?
    Tell 192.168.0.1
  2  0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1?
    Tell 192.168.1.41
  3  0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1?
    Tell 192.168.0.15
  4  0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1?
    Tell 192.168.1.12
  5  0.944186 Apple_68:96:7c → EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1?
    Tell 192.168.0.15
  6  0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.11?
    Tell 192.168.10.113
  7  0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
  8  0.944704 EFMNetworks_55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
  9  0.947778 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
 10  0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
 11  0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6be1c
 12  0.954937 Apple_68:96:7c → Broadcast      ARP 42 ARP Announcement for 192.168.0.15
 13  0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
```

Figure 17: tshark HTTP object export initiated against the decrypted capture (verbose per-packet output shown; extraction itself proceeds silently in the background).

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
1 0.000000 EFMNetworks_55:97:d4 → Broadcast ARP 42 Who has 192.168.0.15?
Tell 192.168.0.1
2 0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.
1? Tell 192.168.1.41
3 0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.
1? Tell 192.168.0.15
4 0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? T
ell 192.168.1.12
5 0.944186 Apple_68:96:7c → EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1
? Tell 192.168.0.15
6 0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10
.1? Tell 192.168.10.113
7 0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transact
ion ID 0x21e6be1c
8 0.944704 EFMNetworks_55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 0
0:26:66:55:97:d4
9 0.947778 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transact
ion ID 0x21e6be1c
10 0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solici
tation from f0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transact
ion ID 0x21e6be1c
12 0.954937 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.16
8.0.15
13 0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solici
tation from f0:f6:1c:68:96:7c
14 0.956994 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.16
8.0.15
```

Figure 18: File-type identification of foremost-carved objects - 2 JPG, 7 GIF (1x1 tracking pixels), 6 PNG, and 2 HTM files.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
ERROR: /home/kali/SBT-DF203-Lab9/exported/foremost is not empty
Please specify another directory or run with -T.

(kali@kali)-[~/SBT-DF203-Lab9]
$ foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
ERROR: /home/kali/SBT-DF203-Lab9/exported/foremost is not empty
Please specify another directory or run with -T.

(kali@kali)-[~/SBT-DF203-Lab9]
$
```

Figure 19: SHA-256 hashing of foremost-carved objects for evidentiary integrity.

9.1 Foremost Carving Results

Type	Count	Notes
JPEG	2	00000104.jpg (2492x2492), 00004538.jpg (718x404, encoder comment "Lavc55.69.100")
GIF	7	All 1x1 pixels — consistent with web tracking/analytics beacons
PNG	6	Sizes ranging 1x1 to 44x44 (UI icons/tracking pixels)
HTM	2	ASCII HTML documents

9.2 HTTP Object Export Results

The HTTP object exporter reconstructed approximately 200 distinct objects directly from completed HTTP transactions, substantially more than foremost's signature-based carving because it uses protocol-aware reassembly rather than raw byte-pattern matching. Notable recovered objects include:

- Images: logo.png, twitter.png, Figure-2.png through Figure-26.png (technical diagrams), ogli5PAh.jpg (718x404 JPEG matching the imgur content viewed), homewave-upwhite.png
- Web assets: jquery-1.10.2.min.js, superfish.min.js, tabs.min.js, bootstrap.min.css, nivo-slider.css, multiple .woff web fonts
- Data: hit.json (7 variants), analytics and tracking payloads referencing scorecardresearch.com, quantserve.com, and amazon-adsystem.com
- HTML: iframe.html, redir.html, ogli5PA.gifv
- Numerous 1x1 GIF tracking pixels associated with third-party advertising and analytics beacons

Many recovered objects share identical SHA-256 hashes despite distinct filenames (e.g. repeated instances of the same tracking-pixel GIF). This is expected behavior for web object extraction: tshark saves a separate file for each HTTP reference to a resource even when the underlying bytes are identical, such as the same 1x1 tracking pixel being requested by multiple page elements. This is not evidence of extraction error or data duplication corruption.

9.3 Reconstructed Browsing Activity

The correlated evidence (Parts E and F) indicates the host device navigated from a Reddit mobile page (m.reddit.com) to an Imgur gallery post titled "Epic Dogpile" (imgur.com/8iVy7t6), which in turn triggered a substantial volume of third-party analytics and advertising traffic (Facebook, Amazon Ad System, ScoreCard Research, Quantserve). One incidental referrer URL captured in an analytics payload referenced an unrelated third-party article on phishme.com; this reflects tracked browsing/referrer metadata swept up by the ad-tech beacons rather than any activity generated during this examination.

10. Part G - Optional Historical Passphrase Challenge

Part G, the optional derivation of the original router passphrase from the recovered 40-bit WEP key via the flawed Microsoft Visual C/C++ linear congruential PRNG (the CodeGate CTF "seed-finding" technique presented in the accompanying lecture module), was not performed as part of this submission. The lab instructions identify this section as optional and instructor-assessed; it was not assigned for this submission.

10.1 Basis for the Optional Challenge (Not Executed)

For completeness, the theoretical basis is noted here without execution: the recovered 40-bit key (A4:3D:F6:F3:74) was originally generated from an ASCII passphrase seeded through a 32-bit PRNG (parameters a=0x000343fd, c=0x269ec3, m=0x00FFFFFF, matching the Microsoft Visual/Quick C/C++ rand() implementation). Recovering the passphrase requires brute-forcing the seed space, deriving candidate passphrases consistent with the known key bytes, and validating candidates against the CTF-supplied SHA-1 hash prefix. This process was not carried out, and no passphrase is reported in this submission.

11. Required Findings Worksheet

Question	Finding
Capture format and duration	pcap, IEEE 802.11 (version 2.4, little-endian), 272.97 seconds, 45,169 packets
BSSID	00:26:66:55:97:D6
ESSID	cgnetwork
Primary station MAC addresses	Host: f0:f6:1c:68:96:7c (Apple); Router: 00:26:66:55:97:d4 / d6
Protected frame count	15,477 WEP data packets
Repeated IV evidence	Yes - IV 0x42e8eb recurred 7 times within a 200-frame sample; several other IVs recurred 4-6 times
Recovered / validated WEP40 key	A4:3D:F6:F3:74 (100% decryption confidence, 83 candidates tested)
Decrypted capture filename / hash	working/file_working-dec, SHA-256 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04
Host IP and MAC	192.168.0.15 / f0:f6:1c:68:96:7c
Server IP and MAC	Local: 192.168.0.9 (port 5000, ASUSTekCOMPU_2a:c2:7a); External: 198.90.20.111, 199.27.79.193
Key protocols observed	DHCP, ARP, DNS, HTTP, TLSv1 / TLSv1.2, IGMPv2, MDNS, ICMPv6
Recovered images / HTML	17 foremost-carved objects (2 JPG, 7 GIF, 6 PNG, 2 HTM) plus approximately 200 objects via tshark HTTP export (images, JS, CSS, fonts, JSON, HTML)
Optional router passphrase	Not attempted - Part G is optional/instructor-assessed and was not performed for this submission
Security conclusion	WEP is cryptographically broken. A 24-bit IV space combined with RC4 keystream reuse allowed near-instantaneous key recovery from 15,477 captured IVs. Recommend migration to WPA2-AES/CCMP or WPA3; legacy WEP should be retired rather than reconfigured.

12. Limitations and Operational Notes

- A HTTP object-export command was run without the tshark -q (quiet) flag, producing a lengthy per-packet console trace during export. This is cosmetic only; the extraction itself completed correctly and files were verified present under exported/http_objects/ afterward.
- An initial foremost invocation failed with "directory is not empty" because exported/foremost had already been created by an earlier mkdir. The directory was removed and foremost was re-run successfully into a clean output path.
- Many carved/exported objects share identical SHA-256 hashes across different filenames. This reflects tshark saving a distinct file per HTTP reference to a resource (e.g. a repeated tracking pixel), not a data-integrity issue.
- Object carving surfaced substantial third-party advertising/analytics traffic and one incidental referrer URL to an unrelated third-party article. These are artifacts of ordinary web browsing and ad-tech beaconing captured within the historical dataset, not activity generated during this examination.
- Part G (optional historical passphrase-cracking challenge) was not performed, consistent with the lab instructions marking that section as optional and instructor-assessed.

13. Evidence Inventory and Reproducibility

Artifact	Role / Contents
evidence/file.xz	Supplied compressed WEP capture; preserved original
working/file_working.xz	Working copy of compressed evidence
working/file_working	Decompressed working capture used for all analysis
working/file_working-dec	Decrypted capture produced by airdecap-ng
reports/file_xz_sha256.txt	Hash of original compressed evidence
reports/working_hashes.txt	Hashes of working compressed and decompressed files
reports/all_working_file_hashes.txt	Hashes of all working directory files post-decryption
reports/capinfos.txt	Capture summary statistics
reports/protocol_hierarchy.txt	Protocol hierarchy breakdown
reports/wlan_frame_sample.tsv	802.11 frame-type sample extraction
reports/wep_protected_frames.tsv	Filtered WEP-protected frame fields including IVs
reports/repeated_iv_summary.txt	IV frequency count demonstrating reuse
reports/aircrack_output.txt	Full aircrack-ng key-recovery output
reports/validated_wep40_key_masked.txt	Recorded recovered key value
reports/airdecap_output.txt	airdecap-ng decryption summary
reports/decrypted_files_inventory.txt	Post-decryption working directory listing
reports/ethernet_endpoints.txt	Ethernet endpoint statistics
reports/ip_endpoints.txt	IPv4 endpoint statistics
reports/tcp_conversations.txt	TCP conversation statistics
reports/ip_mac_mapping_sample.tsv	IP-to-MAC correlation sample
reports/http_export_log.txt	tshark HTTP object export log
reports/foremost_log.txt	foremost carving log
reports/exported_file_types.txt	file(1) type identification of all carved/exported objects

reports/exported_file_hashes.txt	SHA-256 hashes of all carved/exported objects
exported/foremost/	Foremost-carved JPG, GIF, PNG, and HTM files
exported/http_objects/	tshark HTTP-reconstructed objects (images, JS, CSS, fonts, JSON, HTML)

14. Conclusion

The examination successfully reproduced the complete WEP40 attack chain against a historical, authorized training capture: evidence preservation and hashing, 802.11 frame classification, identification of WEP protection and IV reuse, key recovery via aircrack-ng, offline decryption via airdecap-ng, endpoint/conversation analysis, and object carving. Every stage was independently verified through SHA-256 hashing at each transition point, and no live wireless network was scanned, targeted, or connected to at any time.

The recovered 40-bit WEP key (A4:3D:F6:F3:74) was obtained with 100% decryption confidence directly from the captured IVs, requiring only 83 candidate-key tests against 15,477 available IVs. This outcome directly demonstrates the practical weakness of WEP: a 24-bit IV space combined with RC4 keystream reuse renders statistical key recovery trivial on any busy network, regardless of the underlying 40-bit key's nominal strength. Post-decryption analysis further confirmed that once WEP is broken, an attacker gains full visibility into higher-layer traffic - IP addressing, browsing activity, and transferred objects were all fully recoverable from the decrypted capture.

The optional Part G passphrase-derivation challenge was intentionally omitted, consistent with its status as an optional, instructor-assessed exercise. This distinction is preserved so that the submitted findings reflect exactly what was evidenced and executed.

Overall, the completed evidence supports the lab objective of demonstrating WEP's cryptographic inadequacy through a reproducible, fully documented offline forensic workflow, and reinforces the security recommendation that legacy WEP deployments be retired in favor of WPA2-AES/CCMP or WPA3 rather than reconfigured with longer keys.

14.1 Student Declaration

I confirm that I completed this lab in an authorized, isolated environment, preserved the supplied historical evidence, did not target any third-party or live wireless system, and accurately documented my own commands, observations, and conclusions.